

CSA 1717-ARTIFICIAL INTELLIGENCE

LAB EXPERIMENT - 7

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AIM

To develop a Python program that solves the Crypt Arithmetic Problem using a brute-force search by assigning unique digits to letters and finding a valid solution that satisfies the given arithmetic equation (SEND + MORE = MONEY).

Objective, algorithm,result

OBJECTIVE

- To understand the concept of the Crypt Arithmetic Problem in Artificial Intelligence.
- To assign unique digits to letters without repetition.
- To solve the equation $\text{SEND} + \text{MORE} = \text{MONEY}$.
- To verify that the assigned digits satisfy the arithmetic expression.

ALGORITHM

Step 1: Start.

Step 2: Define the letters in the puzzle (S, E, N, D, M, O, R, Y).

Step 3: Generate all possible unique digit assignments (0–9) for these letters.

Step 4: Ensure that the leading letters (S and M) are not assigned zero.

Step 5: Convert the words SEND, MORE, and MONEY into their corresponding numeric values.

Step 6: Check whether the equation $\text{SEND} + \text{MORE} = \text{MONEY}$ is satisfied.

Step 7: If the equation is true, display the solution and the digit assignments.

Step 8: Stop.

SOURCE CODE

```
from itertools import permutations
```

```
letters = ('S', 'E', 'N', 'D', 'M', 'O', 'R', 'Y')
```

```
digits = range(10)
```

```
for perm in permutations(digits, len(letters)):
```

```
    S, E, N, D, M, O, R, Y = perm
```

```
    # Leading digits cannot be zero
```

```
    if S == 0 or M == 0:
```

```
        continue
```

```
    SEND = 1000*S + 100*E + 10*N + D
```

```
    MORE = 1000*M + 100*O + 10*R + E
```

```
    MONEY = 10000*M + 1000*O + 100*N + 10*E + Y
```

```
    if SEND + MORE == MONEY:
```

```
print("Solution Found:")
print("SEND =", SEND)
print("MORE =", MORE)
print("MONEY =", MONEY)
print("\nLetter Assignments:")
print("S =", S)
print("E =", E)
print("N =", N)
print("D =", D)
print("M =", M)
print("O =", O)
print("R =", R)
print("Y =", Y)
break
```

OUTPUT

```
Y
Solution Found:
SEND = 9567
MORE = 1085
MONEY = 10652

Letter Assignments:
S = 9
E = 5
N = 6
D = 7
M = 1
O = 0
R = 8
Y = 2
|
```

RESULT

The Python program successfully solves the Crypt Arithmetic Problem by assigning unique digits to each letter such that the equation $\text{SEND} + \text{MORE} = \text{MONEY}$ is satisfied. The valid solution obtained is:

- $\text{SEND} = 9567$
- $\text{MORE} = 1085$
- $\text{MONEY} = 10652$

Thus, the crypt arithmetic puzzle is solved successfully.